



DIGITAL ASSIGNMENT 5

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Course: Big Data Analytics Lab

Course Code: PMDS507P

Implement and execute a matrix multiplication program using the Hadoop MapReduce framework. The Mapper and Reducer logic will be implemented in Python.

1. Input Data Files

The input consists of two text files, A.txt and B.txt, representing the two matrices. Each line in these files has the format: matrix_name row_index column_index value.

- **Matrix A (A.txt):**
- **Matrix B (B.txt):**



Open ▾ [icon] **B.txt**
~/Desktop/matrixmultiplica...

```
1 B 0 0 5  
2 B 0 1 6  
3 B 1 0 7  
4 B 1 1 8
```



Open ▾ [icon] **A.txt**
~/Desktop/matrixmultiplica...

```
1 A 0 0 1  
2 A 0 1 2  
3 A 1 0 3  
4 A 1 1 4
```

2. Mapper Script (mapper.py)

The Python script mapper.py reads data from standard input. For each element from matrix A, it generates intermediate key-value pairs for each column of the output matrix. For each element from matrix B, it generates pairs for each row of the output matrix. The intermediate key is (i, k), representing the position in the final matrix.

- **Code for mapper.py:**

```
1#!/usr/bin/env python3
2import sys
3for line in sys.stdin:
4    line = line.strip()
5    if not line:
6        continue
7    matrix, row, col, value = line.split()
8    row = int(row)
9    col = int(col)
10   value = float(value)
11
12   n = 2
13
14   if matrix == "A":
15       for k in range(n):
16           print(f"{row}, {k}\tA, {col}, {value}")
17   elif matrix == "B":
18       for i in range(n):
19           print(f"{i}, {col}\tB, {row}, {value}")
20
```

3. Reducer Script (reducer.py)

The script reducer.py receives the grouped key-value pairs. For each key (i, k), it collects the corresponding values from matrix A and matrix B into separate dictionaries. It then calculates the dot product to find the value of the cell and prints the final result.

- **Code for reducer.py:**

```
1#!/usr/bin/env python3
2
3import sys
4from collections import defaultdict
5
6current_key = None
7a_values = defaultdict(float)
8b_values = defaultdict(float)
9
10for line in sys.stdin:
11    line = line.strip()
12    key, value = line.split('\t')
13    if key != current_key:
14        if current_key:
15            result = 0
16            for j in a_values:
17                result += a_values[j]*b_values[j]
18            print(f"{current_key}\t{result}")
19        current_key = key
20        a_values.clear()
21        b_values.clear()
22    tag, index, val = value.split(',')
23    index = int(index)
24    val = float(val)
25
26    if tag == 'A':
27        a_values[index] = val
28    else:
29        b_values[index] = val
30
31if current_key:
32    result = 0
33    for j in a_values:
34        result += a_values[j]*b_values[j]
35    print(f"{current_key}\t {result}")
36
```

4. Preparing HDFS and Scripts

Before running the job, the following preparatory steps were taken:

1. The Mapper and Reducer Python scripts were made executable using the `chmod +x` command.
2. An input directory `/matrix_input` was created in HDFS.
3. The input files A.txt and B.txt were copied from the local filesystem to the `/matrix_input` directory on HDFS.

The directory listing below shows the scripts with execute permissions.

```
hduser@sjt217score008:~/Desktop/matrixmultiplication$ ls -l
total 20
drwxrwxr-x 2 hduser hduser 4096 Oct 17 11:55 input
-rwxrwxr-x 1 hduser hduser 380 Oct 17 12:47 mapper.py
-rwxrwxr-x 1 hduser hduser 380 Oct 17 12:43 mapper.py~
-rwxrwxr-x 1 hduser hduser 711 Oct 17 12:48 reducer.py
-rwxrwxr-x 1 hduser hduser 711 Oct 17 12:43 reducer.py~
hduser@sjt217score008:~/Desktop/matrixmultiplication$
```

5. Executing the Hadoop Job

The Hadoop Streaming utility was used to run the MapReduce job. The command specifies the input and output directories in HDFS, the mapper and reducer scripts, and uses the `-file` option to make the scripts available to the cluster nodes.

```
hduser@sjt217score008:~$ hadoop jar /home/hduser/hadoop/share/hadoop/tools/lib/hadoop-streaming-3.3.1.jar -input /matrix_input -output /matrix_output -mapper /home/hduser/Desktop/matrixmultiplication/mapper.py -reducer /home/hduser/Desktop/matrixmultiplication/reducer.py -file /home/hduser/Desktop/matrixmultiplication/mapper.py -file /home/hduser/Desktop/matrixmultiplication/reducer.py
2025-10-17 12:49:06,410 WARN streaming.StreamJob: -file option is deprecated, please use generic option -files instead.
2025-10-17 12:49:06,448 WARN util.NativeCodeLoader: Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
packageJobJar: [/home/hduser/Desktop/matrixmultiplication/mapper.py, /home/hduser/Desktop/matrixmultiplication/reducer.py] [] /tmp/streamjob3878169385676250640.jar tmpDir=null
2025-10-17 12:49:06,697 INFO impl.MetricsConfig: Loaded properties from hadoop-metrics2.properties
2025-10-17 12:49:06,734 INFO impl.MetricsSystemImpl: Scheduled Metric snapshot period at 10 second(s).
2025-10-17 12:49:06,734 INFO impl.MetricsSystemImpl: JobTracker metrics system started
```

6. Final Output

```
File Input Format Counters
  Bytes Read=64
File Output Format Counters
  Bytes Written=41
2025-10-17 12:49:07,984 INFO streaming.StreamJob: Output directory: /matrix_output
hduser@sjt217score008:~$ hdfs dfs -cat /matrix_output/part-00000
2025-10-17 12:49:46,796 WARN util.NativeCodeLoader: Unable to load native-hadoop library
0, 0    19.0
0, 1    22.0
1, 0    43.0
1, 1    50.0
hduser@sjt217score008:~$
```